

Date: 12-11-2025

1 Objectives

- Matched Filter and BER

2 Tutorial Problems

1. A polar signaling system uses the following pulse in Fig. 1 for transmission with equal probability. It passes through a channel $h(t) = \delta(t)$. Take the noise PSD at the input of the matched filter as $N_0/2$. Draw the matched filter and find signal power, noise power, and BER at the output of the matched filter.

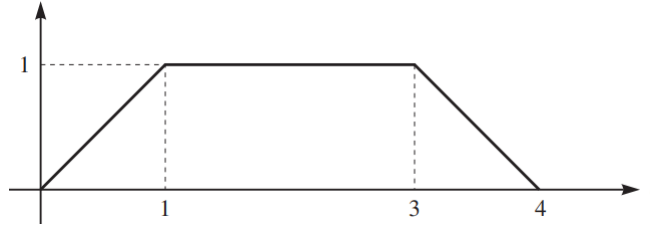


Figure 1: Problem 1

Solution: Denote the given pulse as $p(t)$.

Matched filter $f(t) = p(4 - t) = p(t)$

Amplitude of signal at the output of matched filter = Energy of $p(t)$

$$p(t) = \begin{cases} t, & 0 \leq t \leq 1 \\ 1, & 1 \leq t \leq 3 \\ -t + 4, & 3 \leq t \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

$$E_p = \int_0^1 t^2 dt + \int_1^3 1^2 dt + \int_3^4 (-t + 4)^2 dt$$

$$E_p = \frac{1}{3} + 2 + \frac{1}{3} = \frac{8}{3}$$

$$\text{Amplitude} = \frac{8}{3}$$

$$\boxed{\text{Signal power} = \frac{64}{9}}$$

Noise power at the output of M.F.:

$$P_{n0} = \int_{-\infty}^{\infty} S_n(f) |F(f)|^2 df = \frac{N_0}{2} \int_{-\infty}^{\infty} |F(f)|^2 df$$

$$\Rightarrow P_{n0} = \frac{N_0}{2} \int_0^4 |f(t)|^2 dt = \frac{N_0}{2} E_p = \frac{4N_0}{3}$$

$$\boxed{P_{n0} = \frac{4N_0}{3}}$$

$$\text{BER} = Q\left(\sqrt{\frac{2E_p}{N_0}}\right) = Q\left(\sqrt{\frac{16}{3N_0}}\right)$$

2. BP Lathi Text book question 10-2-1:

10.2-1 In coherent detection of a binary PPM, a half-width pulse $p_0(t)$ of is transmitted with different delays for binary digit “0” and “1” over $0 \leq t \leq T_b$. Note that

$$p_0(t) = u(t) - u(t - T_b/2)$$

The binary PPM transmission is to simply transmit

$$\begin{aligned} & p_0(t), & \text{if “0” is sent} \\ & p_0(t - T_b/2), & \text{if “1” is sent} \end{aligned}$$

The channel noise is AWGN with spectrum level of $\mathcal{N}/2$.

- Determine the optimum receiver architecture for this binary system. Sketch the optimum receiver filter response in the time domain.
- If $P[“0”] = 0.4$ and $P[“1”] = 0.6$, find the optimum threshold and the resulting receiver bit error rate.

(a) Apply Eq. (9.22b) for $q(t) = p_0(t) = u(t) - u(t - \frac{T_b}{2})$ and $p(t) = p_0(t - \frac{T_b}{2}) = u(t - \frac{T_b}{2}) - u(t - T_b)$

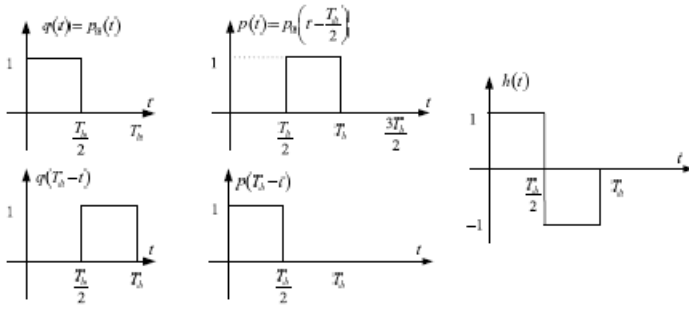


Fig. S9.2-1

$$\begin{aligned} h(t) &= \left[u(t) - u\left(t - \frac{T_b}{2}\right) \right] - \left[u\left(t - \frac{T_b}{2}\right) - u(t - T_b) \right] \\ &= u(t) - 2u\left(t - \frac{T_b}{2}\right) + u(t - T_b) \end{aligned}$$

$$\begin{aligned} E_p &= E_q = \frac{T_b}{2} \\ E_{pq} &= 0 \end{aligned}$$

Hence, the threshold is

$$a_0 = \frac{1}{2} \left(\frac{T_b}{2} - \frac{T_b}{2} \right) = 0$$

(b) The probability of error is calculated by

$$\begin{aligned} P_e &= P[0'] \cdot p_{\text{e|m}}(\epsilon|0) + P[1'] \cdot p_{\text{e|m}}(\epsilon|1) \\ &= P[0'] \cdot Q\left(\frac{a_0 - q(T_b)}{\sqrt{N/2}}\right) + P[1'] \cdot Q\left(\frac{p(T_b) - a_0}{\sqrt{N/2}}\right) \end{aligned}$$

To minimize the P_e , we have

$$\begin{aligned} \frac{\partial P_e}{\partial a_0} &= P[0'] \cdot Q'\left(\frac{a_0 - q(T_b)}{\sqrt{N/2}}\right) \sqrt{\frac{2}{N}} + P[1'] \cdot Q'\left(\frac{p(T_b) - a_0}{\sqrt{N/2}}\right) \sqrt{\frac{2}{N}} \\ &= P[0'] \cdot Q'\left(\frac{a_0 - q(T_b)}{\sqrt{N/2}}\right) \sqrt{\frac{2}{N}} + P[1'] \cdot Q'\left(\frac{p(T_b) - a_0}{\sqrt{N/2}}\right) \sqrt{\frac{2}{N}} \\ &= P[0'] \cdot \frac{1}{\sqrt{\pi N}} \exp\left(-\frac{|a_0 - q(T_b)|^2}{N}\right) \sqrt{\frac{2}{N}} - P[1'] \cdot \frac{1}{\sqrt{\pi N}} \exp\left(-\frac{|p(T_b) - a_0|^2}{N}\right) \sqrt{\frac{2}{N}} \\ &= 0 \end{aligned}$$

or

$$\begin{aligned} P[0'] \exp\left(-\frac{|a_0 - q(T_b)|^2}{N}\right) &= P[1'] \exp\left(-\frac{|p(T_b) - a_0|^2}{N}\right) \\ |a_0 - q(T_b)|^2 - |p(T_b) - a_0|^2 &= N \ln \frac{P[0']}{P[1']} \\ a_0 &= \frac{p(T_b) + q(T_b)}{2} + \frac{1}{p(T_b) - q(T_b)} \frac{N}{2} \ln \frac{P[0']}{P[1']} \\ &= \frac{1}{2} + \frac{N}{2} \ln \frac{2}{3} \end{aligned}$$

3. BP Lathi Text book question 10-6-5:

A source emits M equiprobable messages, which are assigned signals $s_1, s_2, \dots, s_M$, as shown in Fig. 2. Determine the optimum receiver and the corresponding error probability P_{eM} for an AWGN channel as a function of $E_b/\mathcal{N}$.

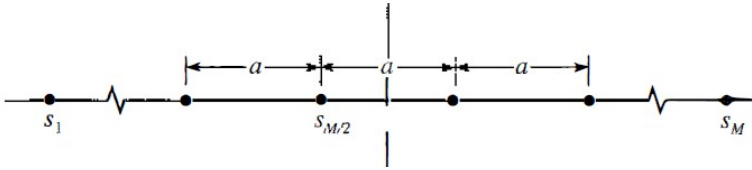


Figure 2: Problem 3

9.6-5 See Figure S9.6-5.

$$P(C|m_1) = \text{Prob}\left(n_1 < \frac{a}{2}\right) \quad \text{and} \quad P(C|m_M) = \text{Prob}\left(n_1 > \frac{-a}{2}\right)$$

$$P(C|m_2) = P(C|m_3) = \dots = P(C|m_{M-1}) = \text{Prob}\left(|n_1| < \frac{a}{2}\right)$$

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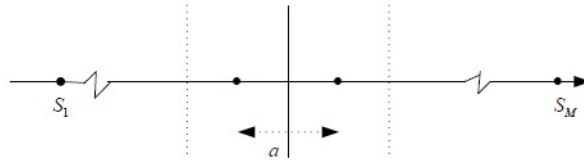


Fig. S9.6-5

Hence

$$P(C|m_1) = P(C|m_M) = \frac{1}{\sqrt{\pi\mathcal{N}}} \int_{-\infty}^{a/2} e^{-x^2/\mathcal{N}} dx = 1 - Q\left(\frac{a}{\sqrt{2\mathcal{N}}}\right)$$

$$P(C|m_2) = P(C|m_3) = \dots = P(C|m_{M-1}) = \frac{1}{\sqrt{\pi\mathcal{N}}} \int_{-a/2}^{a/2} e^{-x^2/\mathcal{N}} dx = 1 - 2Q\left(\frac{a}{\sqrt{2\mathcal{N}}}\right)$$

Hence

$$\begin{aligned} P_{eM} &= 1 - P(C) \\ &= 1 - [P(m_1)P(C|m_1) + P(m_2)P(C|m_2) + \dots + P(m_M)P(C|m_M)] \\ &= 1 - \frac{1}{M} \left[M + 2(M-1)Q\left(\frac{a}{\sqrt{2\mathcal{N}}}\right) \right] \\ &= \frac{2(M-1)}{M} Q\left(\frac{a}{\sqrt{2\mathcal{N}}}\right) \end{aligned}$$

The signal energies are

$$\left(\pm \frac{a}{2}\right)^2; \left(\pm \frac{3a}{2}\right)^2; \dots; \left(\pm \frac{M-1}{2}a\right)^2$$

Hence the average pulse energy $\overline{E}$ is

$$\begin{aligned} \overline{E} &= \frac{2}{M} \left[\frac{a^2}{4} + \frac{9a^2}{4} + \frac{25a^2}{4} + \dots + \frac{(M-1)^2 a^2}{4} \right] \\ &= \frac{a^2}{2M} \sum_{k=0}^{\frac{M-2}{2}} (2k+1)^2 = \frac{(M^2-1)}{12} a^2 \end{aligned}$$

Also

$$E_b = \frac{\overline{E}}{\log_2 M} = \frac{(M^2-1)a^2}{12 \log_2 M}$$

Hence

$$P_{eM} = \frac{2(M-1)}{M} Q\left(\sqrt{\frac{6 \log_2 M}{M^2-1} \cdot \frac{E_b}{\mathcal{N}}}\right)$$